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Paleomagnetism of the Fort Knox Stock, Alaska, and rotation of the Yukon–Tanana terrane after 92.5 Ma

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Abstract

The 92.5 Ma Fort Knox granodiorite stock, near the western end of the Fairbanks Belt in the Yukon–Tanana terrane (YTT) of central Alaska, hosts a world-class gold mine. The stock has been analysed paleomagnetically using thermal and alternating-field step demagnetization and isothermal remanence methods. This pluton retains a primary thermoremanent magnetization at 18 sites (232 specimens) that resides mainly in single-to pseudosingle-domain magnetite with a direction of $D=228.8°$, $I=84.3°$ ($N=18$, $k=130$, $\alpha_{95}=3.0°$), giving a paleopole at 56.5°N, 197.1°E ($dp=5.9°$, $dm=5.8°$). The pluton's host rock, the Fairbanks schist, does not retain a stable coherent remanence. Relative to the North American craton, the stock's paleoinclination indicates that the Fairbanks Belt has undergone nonsignificant poleward (northwesterly) translation of 25±750 km only. Analysed in concert with the few available paleoinclinations available for the YTT in Yukon, the paleoinclination suggests further that the YTT has undergone only ~250 to 450 km of dextral displacement along the Tintina fault in the past ~100 Ma and, therefore, is parautocthonous since the mid-Cretaceous. The stock's paleodeclination records 121±35° of counterclockwise rotation relative to the North American craton. Consideration of models published for Alaska's tectonic evolution suggests that this paleodeclination discordance is caused by rotations associated with the opening of the Canada Basin, with dextral displacement on the Tintina fault, and with development of the western Alaskan orocline. Thus the paleomagnetic results for the Fort Knox stock support a thin-skin tectonic model for the accretion of the YTT and Intermontane Belt terranes to the northern Cordillera.
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Keywords: Paleomagnetism; Fort Knox; Alaska; Tectonics; Yukon–Tanana terrane

1. Introduction

A score of paleomagnetic studies sponsored by the Lithoprobe SNORCLE (Slave-NORthern Cordillera Lithospheric Evolution) transect project have investigated the geotectonic evolution of the Intermontane Belt and Yukon–Tanana terranes (Symons et al., 2005). In this paper we study the paleomagnetism of the mid-Cretaceous Fort Knox stock, which is of particular interest because it is located near the western end of the Yukon–Tanana terrane (YTT; Fig. 1a) and thus may have been involved in the postulated Late Cretaceous–Paleocene counterclockwise rotation of Alaska, first proposed by Carey (1955, 1958) (Fig. 1b), as well as in several other geotectonic events since Early Cretaceous time (e.g., Plafker and Berg, 1994a,b). For example, based on similar early Paleozoic to mid Cretaceous lithologic sequences, the YTT may extend southwest of

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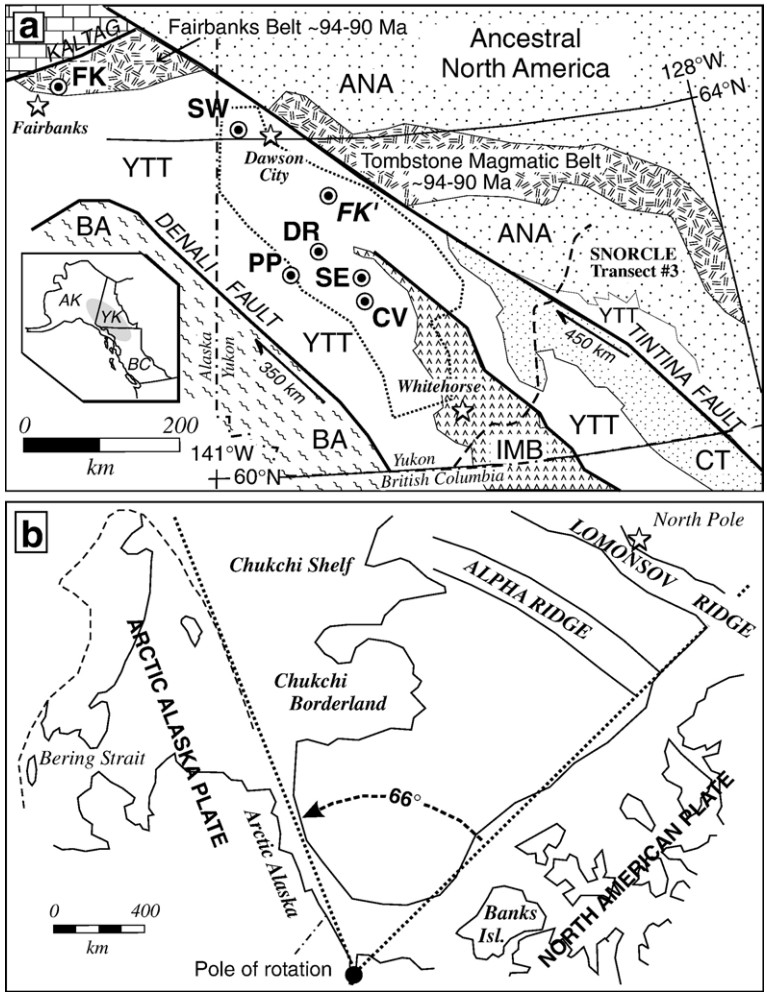


Fig. 1. (a) Locations of paleomagnetic collections in the Yukon–Tanana terrane (YTT) and Fairbanks–Tombstone (–Tungsten) magmatic belt, adapted from Selby et al. (2003). ANA, Ancestral North America; BA, Baja Alaska; BC, Big Creek batholith; CV, Carmacks volcanics; DR, Dawson Range batholith; FK, Fort Knox pluton; FK′, emplacement location of the Fort Knox pluton relative to North America from the reconstruction of Plafker and Berg (1994a,b); GM, Granite Mountain batholith; IMB, Intermontane Belt terranes; INS, Insular Belt terranes; PP, Pattison pluton; SE, Seymour Creek stock; SW, Swede Dome stock. The references for the collections are given in Table 2. (b) Opening of the Canada Basin, adapted from Grantz et al. (1979).

Fairbanks as the Dillinger terrane, then northeast as the Ruby terrane, and then westerly as the Arctic Alaska terrane (Box, 1985; Dover, 1994; Johnston, 2001), in which case the northern half of Alaska may have undergone extreme tectonism after the mid-Cretaceous. In this study we compare the paleomagnetic result for the Fort Knox stock with the coeval Cretaceous North American cratonic reference pole, and also with coeval data from Arctic Alaska, the more southerly terranes in Alaska, and the accreted Intermontane Belt terranes of Yukon and British Columbia. The Fort Knox stock provides the first Cretaceous paleomagnetic result from the western YTT, thus allowing an assessment of the YTT's role in various proposed geotectonic options for the Alaskan and Yukon region.

2. Fort Knox pluton

The Fort Knox pluton (i.e., the Vogt stock) is located about 22 km northeast of Fairbanks, Alaska, in the northwestern part of the YTT (Bakke, 1995; Fig. 1a). The pluton is emplaced into the Fairbanks schist, a polymetamorphic micaceous-quartz schist and mica quartzite of Precambrian to mid-Paleozoic sedimentary origin. Ovate in shape – about 1100 m long by 600 m wide – this composite pluton is composed of three major

phases: an early biotite-rich fine-grained granite, followed by a medium-grained porphyritic granite, and a younger coarse-grained seriate porphyritic granite. These granites grade towards granodiorite in composition. The contact between the granite pluton and the host schist plunges steeply away from the pluton to the east and moderately away to the north, west and south.

The Fort Knox pluton hosts a world-class gold deposit (Bakke, 1995). Gold–bismuth–tellurium sulphide mineralization at Fort Knox occurs only in the pluton where it is associated with a stockwork of sheeted veins. In detail, the mineralization, which constitutes only ~0.1% by volume, is found both in and along the margins of narrow pegmatite veins, quartz veins and veinlets, shear zones, and fractures. Fluid inclusions in the pegmatite and quartz veins suggest that the fluid was trapped at pressures of 1.25 to 1.50 kbar or ~5 km or less depth and at temperatures of ~300±30 °C (Bakke, 1995). The sulphide minerals include pyrrhotite (Thompson et al., 1999). This deposit contains ~158 Mt at 0.83 g/t Au, is mined in an open pit, and is the premier gold producer in Alaska (Thompson et al., 1999).

The ages of the pluton and its mineralization are tightly constrained. U/Pb zircon dating of the granite has given an age of 92.5±0.2 Ma (Bakke, 1995; J.K. Mortensen, personal communication in Selby et al., 2002). Within error, this age is identical to the average of six $^{187}Re/^{187}Os$ ages of 92.6±0.9 Ma determined from molybdemite for the mineralization (Selby et al., 2002). Given an emplacement depth of ~5 km for the Fort Knox deposit (Lang et al., 2000), relatively rapid cooling of this small high-level pluton to <200 °C is likely. Rapid cooling is also consistent with $^{40}Ar/^{39}Ar$ muscovite plateau ages of 87.4±0.3 Ma for pegmatite and of 88.1±0.4, 87.9±0.4 and 86.8±0.4 Ma for veins (McCoy et al., 1997), which tend to be too young by a few million years in 92.5 Ma rocks (Dunlap, 2000). Two zircon fission-track ages of ~51 Ma and four apatite fission-track ages of ~42±6 Ma (Murphy and Bakke, 1993) record later heating events of <200 °C. The ~50 Ma event is thought to be related to the extrusion of the ~50 Ma Eocene basalts found at six localities in the Fairbanks Mining District that have reset some $^{40}Ar/^{39}Ar$ ages in the area (Roe and Stone, 1993; Douglas et al., 2002).

Regionally, the Fort Knox stock belongs to the 90 to 94 Ma Fairbanks–Salcha plutonic suite of peraluminous porphyritic granodiorite–granite plutons in the YTT (Fig. 1) (Hart et al., 2004). The suite outcrops in an ~150-km-long belt eastwards from Fairbanks, Alaska. The Fairbanks–Salcha plutons have a low magnetite content (Thompson et al., 1999; Hart et al., 2004), and so give a uniformly low response in ground magnetometer surveys (Bakke, 1995).

3. Sampling

Six cores, oriented by sun compass, were collected from each of the 22 sites in the Fort Knox open pit (Fig. 2). Of these 22 sites, 18 were collected in granite of the Fort Knox pluton, three in the host rock schist, and one in a one-metre wide pegmatitic dike. From all cores, 281 specimens were prepared, each 2.54 cm in diameter and 2.20 cm long. Next the specimens' magnetic bulk susceptibilities were measured on a Sapphire Instruments SI-1 meter. The 243 granitic specimens had a median bulk magnetic susceptibility of 117.8 SI (quartile values: Q_1=95.2 SI, Q_3=167.1 SI), which was notably lower than that of the 38 schist specimens (224.1 SI;

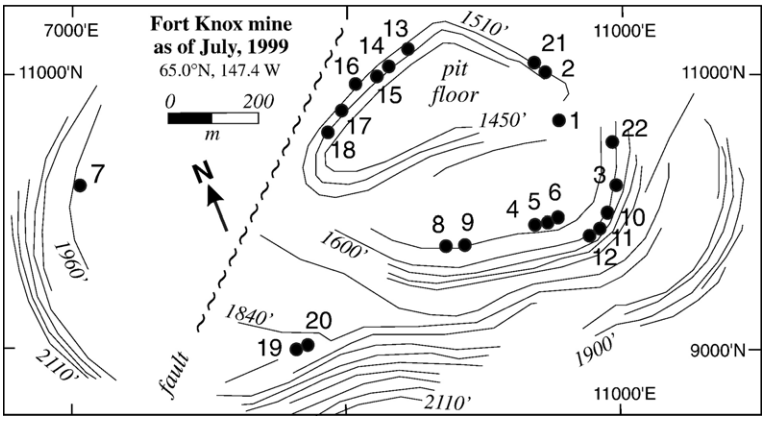


Fig. 2. Simplified bench map of the Fort Knox open pit mine as of July, 1999, with locations of the paleomagnetic sampling sites. Some bench elevations in feet are noted.

Q_1=138.3 SI, Q_3=299.8 SI), but is consistent with the pluton generating a magnetic low anomaly. The specimens were then stored for several months in a magnetically-shielded room with an ambient field of ~100γ to allow their viscous remanent magnetizations to decay. The specimens were kept in the room until all subsequent measurements were completed.

4. Remanence measurements

The natural remanent magnetization (NRM) of each specimen was measured on an automated vertical-axis 2G Enterprises DC-SQUID magnetometer with a sensitivity limit of ~2×10^{-6} A/m. The 243 granitic specimens had a median intensity of 8.8×10^{-4} A/m ($Q_1=5.4\times10^{-4}$ A/m, $Q_3=16.8\times10^{-4}$ A/m) and the 38 schist specimens had a somewhat lower value of 4.4×10^{-4} A/m ($Q_1=2.4\times10^{-4}$ A/m, $Q_3=6.3\times10^{-4}$ A/m).

Two pilot specimens with "average" NRM intensities and directions were selected from each site and subjected to alternating-field (AF) and thermal (TH) step demagnetization tests. AF demagnetization was done in 14 steps to 120 mT using a Sapphire Instruments SI-4 demagnetizer and TH demagnetization was done in 16 steps to 600 °C using a Magnetic Measurements MM-1 demagnetizer. After determining the optimum demagnetization treatment to isolate the stable characteristic remanent magnetization (ChRM) at each site, most of the remaining specimens were TH demagnetized and the residual specimens AF demagnetized in 8 or more steps. Also 10 specimens, representative of the lithologies and ChRM behaviours, were magnetized in 13 steps up to 900 mT using a Sapphire Instruments SI-6 DC pulse magnetizer to define their acquisition of saturation isothermal remanence (SIRM), and then they were AF demagnetized in 10 steps to 160 mT to define the decay of their SIRM.

TH step demagnetization data showed that the ChRM in 11 granitic sites was carried by magnetite (Fig. 3a), and in the other 8 sites by both pyrrhotite and magnetite (Fig. 3b). When the TH results for 8 granite specimens were matched against their SIRM crossover points for the same sites (Symons and Cioppa, 2000), the ChRM was found to be carried by single domains in both magnetite and pyrrhotite, giving crossovers with mean values of 50±14 mT and 0.43±0.03 J/J_{900}, respectively. In general, AF step demagnetization proved less effective at isolating the ChRM component direction.

The specimen ChRM directions were isolated using both vector component analysis (Kirschvink, 1980) and remagnetization circle analysis (Bailey and Halls, 1984), and site mean directions were obtained using

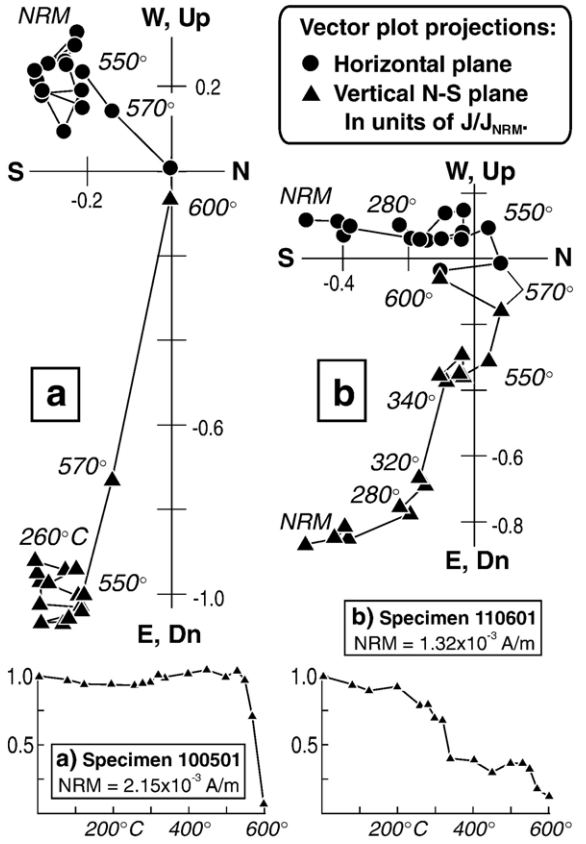


Fig. 3. Vector component plots for two example specimens of granite on thermal step demagnetization showing the isolation of: (a) a magnetite-borne ChRM component; and (b) a pyrrhotite and magnetite-borne ChRM component. The plot axes are: N — north, E — east, S — south, W — west, Up — upwards, and Dn — downwards. Values on the axes are expressed as a ratio of the natural remanent magnetization (NRM) intensities of 2.15 mA/m in "a" and 1.32 mA/m in "b." Below each component plot is a plot of the intensity ratio against demagnetizing temperature. The rapid intensity drop between 280 and 340 °C is indicative of pyrrhotite and that between 550 and 600 °C is indicative of magnetite.

Fisher (1953) analysis (Table 1). The granite stock sites gave vertical to steeply-downward southwesterly ChRM mean directions (Fig. 4a). Most of the stock's sites have well-clustered ChRM directions with 95% confidence radii of $\alpha_{95}<10°$. The mean ChRM direction is much better defined in magnetite than in pyrrhotite. Site 8 gave a divergent and more-scattered mean ChRM direction relative to the other sites, and so it was omitted in calculating the Fort Knox stock's mean ChRM direction of declination (D)=228.8°, inclination (I)=84.3° (N=18, k=130, α_{95}=3.0°; number of site means, precision parameter of Fisher (1953), and radius of cone of 95% confidence about the site mean). The ChRM is deemed to be a primary themoremanent

Table 1
Fort Knox pluton: site mean remanence directions

Site	Specimens			ChRM direction				α_{95} (°)
	e	*r*	*o*	*Dec* (°)	*Inc* (°)	*n*	*k*	
01	9	2		6.5	86.0	11	22.8	9.8
03	12			190.8	81.2	12	77.5	5.0
04	12			229.5	72.0	12	190.3	3.2
05	5	6	1	248.9	76.3	11	70.9	5.5
06	10		1	275.8	81.9	10	61.9	6.2
08*	11		1	188.5	60.6	11	21.1	10.2
09	7	4	1	213.5	78.5	11	46.9	6.7
10	12			211.6	80.5	12	506.6	1.9
11	12			145.4	88.8	12	44.2	6.6
12	10	2		291.3	84.5	12	67.7	5.3
13	12	1		195.3	83.2	13	58.5	5.5
14	12		1	181.5	80.9	12	55.7	5.9
15	12			206.1	87.1	12	28.8	8.2
16	8	5	1	54.8	85.8	13	36.4	7.0
17	14			210.4	77.3	14	55.7	5.4
18	8	2	1	180.6	86.5	10	39.1	7.8
19	11	2		285.3	80.0	13	31.7	7.5
21	5	6		322.0	88.4	11	46.3	6.8
22	12			245.5	80.5	12	115.0	4.1
18 sites*				228.8	84.3	18	129.7	3.0

Notes: *Specimens*: *e* — endpoints, *r* — remagnetization circles, *o* — omitted; *ChRM direction: Dec* — declination in degrees, *Inc* — inclination in degrees, *n* — number of vectors used, *k* — precision parameter estimate (Fisher, 1953), α_{95} — radius of cone of 95% confidence. * — site 8 mean direction omitted from the unit average as aberrant.

magnetization for several reasons. First, the pluton provides no petrologic evidence of post-mineralization secondary alteration resulting from a chemical or thermal overprint event. Second, the pluton is massive, militating against a dynamic overprinting of the ChRM. Third, remagnetization would have moved the paleopole back onto the North American apparent polar wander path (APWP) at some time after 92 Ma, and this is not the case. Fourth, had the Fort Knox been remagnetized after 92 Ma, then surrounding pyrrhotite-bearing host rocks would have been remagnetized with the same ChRM direction, and this did not happen. Fifth, any remagnetization event that would have changed the ChRM direction would have likely occurred after the 118 to 84 Ma Cretaceous normal superchron (Opdyke and Channell, 1996) so that some reversed polarity directions would be expected and none were found. The resulting paleomagnetic pole position for the ChRM of the Fort Knox stock (site location: 65.0°N, 212.6°E) is at 56.5°N, 197.1°E (dp=5.9°, dm=5.8°; radii of oval of 95% confidence). In contrast to the granite, sites in the host rock schist gave randomly scattered and uninterpretable specimen ChRM directions that resided in pyrrhotite only (Fig. 4b).

5. Tectonic discussion

5.1. Yukon Tanana terrane

In east central Alaska and Yukon the definition of the terranes and subterranes bounding the YTT is continually evolving (Plafker and Berg, 1994a,b). Except for slivers of the Windy, McKinley and Pingston terranes, the YTT is bounded to the southwest by the Denali fault with the accreted terranes of southern or Baja Alaska on the other side i.e., the Peninsular–Alexander–Wrangellia superterrane in southern Alaska excluding the Alaskan panhandle (Packer and Stone, 1974) (Fig. 1). The Tintina fault

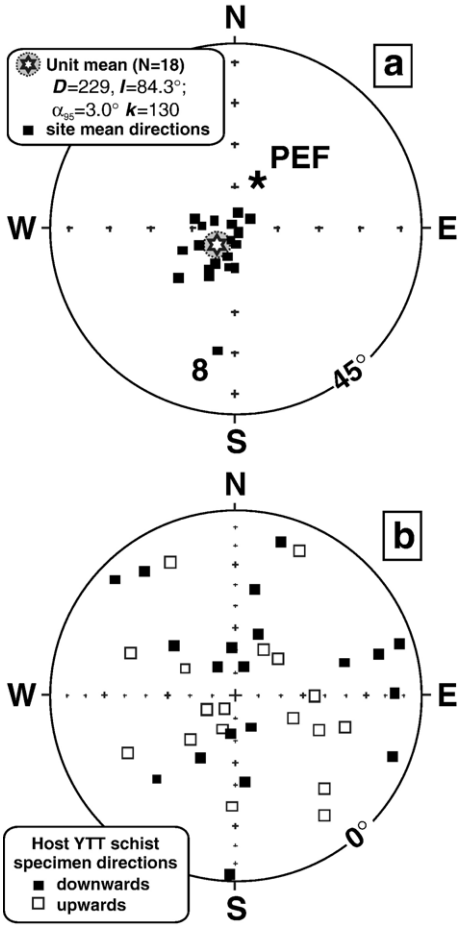


Fig. 4. (a) Central portion (>45° inclination) of the lower hemisphere of an equal-area stereonet showing site mean directions (solid squares) for the Fort Knox stock along with their mean direction (star) and its cone of 95% confidence (shaded circle), and the present-day Earth's magnetic field (PEF) direction at the Fort Knox location. (b) Equal-area full stereonet showing random specimen ChRM directions for three sites in the host Fairbanks schist. Solid (open) squares denote downward (upward) vector directions.

marks the northeast side of the YTT in Yukon with the Selwyn fold and thrust belt of North America on the other side, except for a slice of the YTT on the northeast side of the fault in southeastern Yukon. North of the Kaltag fault and northwest of the Tintina fault in Alaska is the Arctic Alaska terrane, dominated by the east–west trending Brooks Range with the Colville basin to the north. Extension of the YTT to the west–southwest beyond Fairbanks around the western Alaskan orocline (Plafker and Berg, 1994a,b) is problematic with the area between the Kaltag and Farewell faults assigned to the Koyukuk, Minchumina, Nixon Fork, Ruby and other terranes. Finally, at depth, Lithoprobe SNORCLE refraction and near-vertical and wide-angle reflection seismic profiles in Yukon and Northwest Territories indicate that the YTT either rests directly on a westward-thinning ramp of ancestral North American Proterozoic strata (1.85 to 0.54 Ga) or rests on Paleozoic strata that rests in turn on the Proterozoic strata (Snyder et al., 2002; Cook et al., 2004; Hammer and Clowes, 2004). The ramp extends westward almost to the Denali fault, and its western edge is interpreted to be the rifted Proterozoic continental margin of the North American craton. Thus the YTT and Intermontane Belt (IMB) terranes are interpreted to be thin suspect terranes that were thrust onto and up the ramp from the southwest on sole faults as upper-crustal "flakes" by thin-skin tectonics (Symons et al., 2000; Snyder et al., 2002). Whether or not the YTT in Alaska is grounded on a cratonic ramp remains unknown. In Yukon, the southeastern end of the YTT is covered by strata of the IMB terranes.

Compared to North America, the YTT has a different and distinctive Devonian through early Jurassic lithologic sequence. Based on geologic evidence, therefore, the accretion of the YTT to North America is thought to have occurred in Early to Middle Jurassic (e.g. Tempelman-Kluit, 1976; Mortensen and Jilson, 1985; Mortensen, 1992; Hansen and Dusel-Bacon, 1998; Cook et al., 2004). In contrast, the paleomagnetic evidence has been interpreted in two ways. One alternative suggests that the YTT has been pericratonic or parautochthonous with North America since the Jurassic (e.g. Irving et al., 1995; Symons et al., 2005), and the second alternative suggests that the YTT was an independent microplate that was translated 3000 ±500 km northward with the IMB terranes between ~85 Ma and ~45 Ma, at which time it accreted to North America (e.g. Johnston et al., 1996; Johnston, 2001).

In the Jurassic and Early Cretaceous, the YTT underwent contraction, developing northwest-striking structures (e.g. Gabrielse et al., 1991; Mihalynuk et al., 1994). From 115 to 90 Ma, 8 extensive plutonic suites were emplaced throughout the YTT, of which the 94–90 Ma Fairbanks–Salcha suite is one of the youngest (Mortensen et al., 2000; Hart et al., 2004). Coincidentally, the YTT underwent episodes of extension and dextral strike-slip faulting (Mortensen, 1992; Hansen and Dusel-Bacon, 1998). Based on the areal exposure of intrusive and volcanic rocks, only minor plutonism occurred in the YTT from ~90 Ma to the present (Mortensen, 1992; Pavlis et al., 1993). However, active dextral strike-slip faulting did occur in the YTT, resulting in hundreds of kilometres of displacement in total (Gabrielse et al., 1991). Most notably, the correlation of the 90–92 Ma Livengood plutonic suite in Alaska about 100 km north of Fairbanks to the coeval Tombstone suite in Yukon, as well as other correlations, records ~425 km of dextral displacement on the Tintina fault alone (Gabrielse, 1985; Gabrielse et al., 1991; Gordey and Makepeace, 2003; Hart et al., 2004).

5.2. Early tectonic models for Alaska

Carey (1955) speculated that Alaska rotated ~30° counterclockwise (CCW) to open a sphenochasm, the Canada Basin, in the Arctic Ocean as the North Atlantic opened (Fig. 1b). Packer and Stone (1972) reported paleomagnetic data from Jurassic rocks from three areas in southern Alaska to show, on the contrary, that the host terranes had been rotated clockwise (CW) and derived from more southerly paleolatitudes, leading to the Baja Alaska hypothesis (Packer and Stone, 1974) and subsequently to the Baja British Columbia (Baja BC) hypothesis (Irving, 1985). Many paleomagnetic studies were undertaken during the 1970s and 1980s to test various hypotheses for the tectonic evolution of the several Alaskan terranes as summarized by Hillhouse and Coe (1994). Until recently, however, little paleomagnetic attention had been paid to the YTT, and no paleomagnetic results have been reported from it in Alaska.

Most Late Cretaceous and younger rocks from Baja Alaska systematically record both substantial CCW rotations and northwards transport from much more southerly paleolatitudes (Harbert et al., 1990). Hillhouse and Coe (1994) concluded that the Wrangellia and Peninsular terranes collided with the southern sides of the Nixon Fork and Yukon–Tanana terranes between ~100 Ma and 65±10 Ma, having been carried northward on the Kula plate. Thereafter, the Chugach, Prince William and Yakutat terranes accreted on the southern shore to form Baja Alaska. Also they concluded that the western portion of Baja Alaska rotated 30° to 50° CCW between 68 and 44 Ma about a hinge at ~214°E longitude near the western end of the YTT to form the western Alaskan orocline.

Paleomagnetic data from Arctic Alaska led both Stone (1989) and Harbert et al. (1990) to conclude that, despite extensive remagnetization overprints from burial metamorphism or orogenic fluid flow at most sites, the few reliable paleomagnetic sites recorded CCW rotations consistent with Carey's (1955) rotation model for Alaska. Stone (1989) also observed, in contrast, that sites in the Minchumina and Nixon Fork terranes recorded substantial CW rotations and invoked the "ball-bearing" hypothesis of Beck (1980) to explain them. Harbert et al. (1990) deemed the paleopole from well cores of the Early Cretaceous (Neocomian) Kuparuk River Formation (Halgedahl and Jarrard, 1987) of the Alaskan North Slope to be primary. Further, they deemed the mid-Cretaceous Nanushuk Formation paleopole (Witte, 1982; Witte et al., 1987) and other paleopoles from older rock units in Arctic Alaska to have been remagnetized during the mid-Cretaceous Brookian Orogeny. Thus Harbert et al. (1990) concluded that Arctic Alaska had been rotated ~66° CCW away from the Arctic islands of Canada at the edge of the North American craton about an Euler pole in the Mackenzie River delta to open the Canada Basin between ~136 Ma and ~95 Ma, as suggested by Tailleur (1969) and Grantz et al. (1979, 1990).

A recent paleomagnetic study of the ~160 Ma Asik Mountain mafic complex in the Brooks Range of Arctic Alaska (Lewchuk et al., 2004a) is relevant to this discussion. From anisotropy of magnetic susceptibility and radiometric age dating data, Lewchuk et al. (2004a) showed that the complex's ChRM is primary, and concluded from its paleopole that Arctic Alaska has undergone ~70° of CCW rotation with possibly a statistically-nonsignificant northward translation of ≤500 km. Also Lewchuk et al. (2004b) have done a paleomagnetic study of the Red Dog Zn–Pb massive sulphide deposit in the western Brooks Range of Arctic Alaska. The deposit carries two magnetizations. ChRM A is carried by magnetite, is found in rocks with barite and/or quartz replacement, fails the paleomagnetic fold test, and is interpreted to be an 100 to 150 Ma remagnetization caused by hydrothermal activity. A variety of tectonic explanations are possible to explain the discordant paleopole from ChRM A depending on the chosen location for the Euler pole of rotation and on the age assumptions, however, the simplest interpretation is an ~70° CCW rotation, which is the same as for the Asik Mountain complex given above. ChRM B from the Red Dog deposit is carried by pyrrhotite and is found in ores rich in galena and sphalerite that have been dated radiometerically at 338 Ma by Re–Os on pyrite. Lewchuk et al. (2004b) noted that one valid interpretation of many possibilities was that Arctic Alaska may been translated northward by ~30° relative to North America before 160 Ma and then rotated CCW by 50° to 70° after 160 Ma.

Tectonic models for the formation of the Canada Basin have been reviewed by Lawver and Scotese (1990) who noted that the models fall into three categories: oceanization of continental crust, entrapment of old oceanic crust, and the in situ formation of oceanic crust by seafloor spreading. Of these, only the lattermost category can explain the available geological and geophysical evidence. Lawver and Scotese (1990) noted four distinct variants of seafloor spreading models. The Canada Basin (Fig. 1b) is opened in: variant "a," by the rotation of Arctic Alaska away from the North American craton as Carey (1958) proposed; variant "b," by translation of Arctic Alaska southwestward from the Alpha or Lomosov ridge; variant "c," by translating the Chukchi borderland westward past an unmoving Arctic Alaska; and variant "d," by north–northwestward translation of Arctic Alaska as part of the craton along the Tintina fault and its presumed northwestern extension. The proposed timing for the opening of the Canada Basin varies, i.e., ~119 to 97 Ma (Tailleur and Brosgé, 1970), ~125 to ~80 Ma (Lawver and Baggeroer, 1983), ~130 to ~100 Ma (Halgedahl and Jarrard, 1987), ~135 to ~70 Ma (Grantz et al., 1998), and ~175 to ~120 Ma (Lawver et al., 2002).

5.3. Post-mid-Cretaceous YTT displacement

Using the 20 Ma moving-average window data for the North American apparent polar wander path (APWP) of Besse and Courtillot (2002) to obtain the 92.5 Ma reference pole for the craton, the Fort Knox stock gives a significant (2σ) calculated rotation of 121°±35° CCW and a nonsignificant poleward translation of 0.2°±6.4° (25±720 km) using the method of Demarest (1983). On its own this paleolatitudinal result suggests strongly that the YTT has not been translated northward or southward along the cratonic margin following emplacement at 92.5 Ma.

Adding in the Fort Knox paleopole, there are now six 56 to 96 Ma paleopoles reported for the YTT (Table 2). By comparing them to their coeval cratonic reference paleopoles, paleolatitudinal displacement estimates can be calculated to obtain northward (poleward) or southward translation estimates along the cratonic margin. The six units have an average age of 76±14 Ma and indicate a nonsignificant northward translation of 5.9°±8.5° or 650±950 km. Excluding the apparently anomalous translation value for the Carmacks volcanics (Table 2) increases the average age to 77±15 Ma and reduces the northward translation estimate to a nonsignificant 2.4°±3.4° or 265±375 km. Given that Late

Table 2
Net tectonic motion estimates from <100 Ma paleopoles for the Yukon–Tanana Terrane[a]

Pole source, unit abbreviation		Age	Sites	Location (°)		Paleopole (°)			Translation	Rotation	Ref.[a]
		Ma		Lat. N	Long. E	Lat. N	Long. E	α_{95}	Poleward [b]	Clockwise	
Dawson Range batholith	DR	96±4	11	62.6	221.6	80.3	221.7	7.2	1.9±7.7°	20±28°	2
Fort Knox pluton	FK	93±1	18	65.0	212.6	56.5	197.1	5.9	0.2±6.4°	−121±35°	1
Carmacks Gp. volcanics	CV	71±2	26	62.1	222.8	78.4	088.6	7.8	23.1±7.0°	−6±17°	3
Swede Dome stock	SW	70±1	18	64.2	219.6	77.3	224.7	6.0	2.0±5.6°	25±26°	4
Seymour Creek stock	SE	69±1	16	62.3	222.8	55.2	202.5	8.4	−0.9±7.3°	−94±36°	5
Pattison pluton	PP	56±1	13	62.4	222.1	86.6	148.1	8.0	8.9±6.9°	17±16°	4

[a]Motion estimates are calculated with respect to the 10 Ma (20 Ma smoothing window) paleopoles of Besse and Courtillot (2002; their Table 4) for North America, interpolated as follows: *96 Ma*: 76.4°N, 198.2°E, α_{95}=6.4°; *93 Ma*: 76.3°N, 203.5°E, α_{95}=5.5°; *71 Ma*: 74.4°N, 205.6°E, α_{95}=3.6°; *70 Ma*: 74.3°N, 205.2°E, α_{95}=3.6°; *69 Ma*: 74.3°N, 205.1°E, α_{95}=3.5°; *56 Ma*: 76.8°N, 188.9°E, α_{95}=3.1°.
[a]References are: 1 — this study; 2 — McCausland et al. (in press); 3 — Wynne et al. (1998); 4 — McCausland (2002); 5 — McCausland et al. (2001).
[b]One degree of poleward transport is equal to ~111 km.

Cretaceous to mid-Eocene displacement of the YTT would be northward if the YTT was part of Baja BC, the fact that the "average" age of the units is ~77 Ma suggests that the 2.4°±3.4° estimate is likely to be a couple of degrees too low. However, given that the sampling sites are close to the reference poles, the estimates do not allow for dispersion of the paleopoles about their reference poles and hence the estimates are likely to be too high by an offsetting couple of degrees. Although not rigorous, the data suggest that the YTT has moved northward by only ~265±375 km relative to North America during the Late Cretaceous and Paleocene. Geologic estimates of the northward displacement of the YTT on the Tintina Fault zone relative to North America have varied, but most have been on the order of ~425 km, as noted earlier in this paper. This agreement leads to the conclusion that the YTT was pericratonic or parautocthonous with respect to North America since the emplacement of the Fort Knox stock at 92.5 Ma, implying that the YTT was not translated significantly in paleomagnetic terms along with either Baja BC or Baja Alaska, and that the CCW rotation found for the Fort Knox stock must be explained by rotation around a nearby axis located within a very few hundred kilometres of the pluton.

5.4. Rotation

Several explanations merit consideration to explain Fort Knox's deviant paleopole, i.e., local tilt, local rotation, rotation with Arctic Alaska, rotation in response to dextral strike-slip motion on the Tintina fault system, or some combination of rotations.

Post-emplacement tilt of a small pluton may be recognized or suspected in various ways such as the distribution of petrologic phases, attitude of the host rocks, foliation attitudes, contact attitudes from deep drilling, analysis of magnetic or gravity anomalies, etc. The Fort Knox pluton is massive, was emplaced into highly deformed metamorphosed host rocks, and provides no evidence of any post-emplacement tilt (P. Jensen, personal communication, 2003). Furthermore, there are good reasons to think that a local tilt or rotation is unlikely. If a pluton has been tilted in an orogenic belt, then the tilt axis tends to be parallel to the structural grain of the host rocks surrounding the pluton. To move the 92.5 Ma reference paleopole to the observed Fort Knox paleopole requires a tilt of ~10±3° up of the pluton's southwest side about a horizontal axis trending 110–290°. Such an east–southeasterly tilt axis trend is dissimilar to the southwesterly trend observed for the pluton and the adjacent regional geology (Robinson et al., 1990; Newberry et al., 1996). Nevertheless, the possibility of an unrecognized local postemplacement tilt of the Fort Knox stock to give its discordant pole position cannot be entirely ruled out. Similarly, there is no local deformational evidence such as bounding adjacent sinistral transcurrent faults to suggest an ~121° CCW vertical-axis rotation of the pluton. These local relationships suggest that the rotation involved the region at either the westernmost end of the Fairbanks belt only or of all of the YTT west of the Tintina Fault zone.

The third option is that the Fort Knox region was rotated along with Arctic Alaska following the spenochasm model of Carey (1955). Using a best fit given by the 1000 m isobath (Grantz et al., 1979), the opening of the Canada Basin predicts a rotation of 66° CCW away from the North American craton about a vertical axis located in the Mackenzie River delta at 69.1°N, 229.5°E. Correcting the Fort Knox paleopole for this rotation moves it on a small circle into much better agreement with the cratonic reference paleopole (Fig. 5a). In Fig. 5a, note nevertheless that the 2σ cones of confidence for the corrected Fort Knox paleopole and the corresponding cratonic reference pole barely intersect, showing with

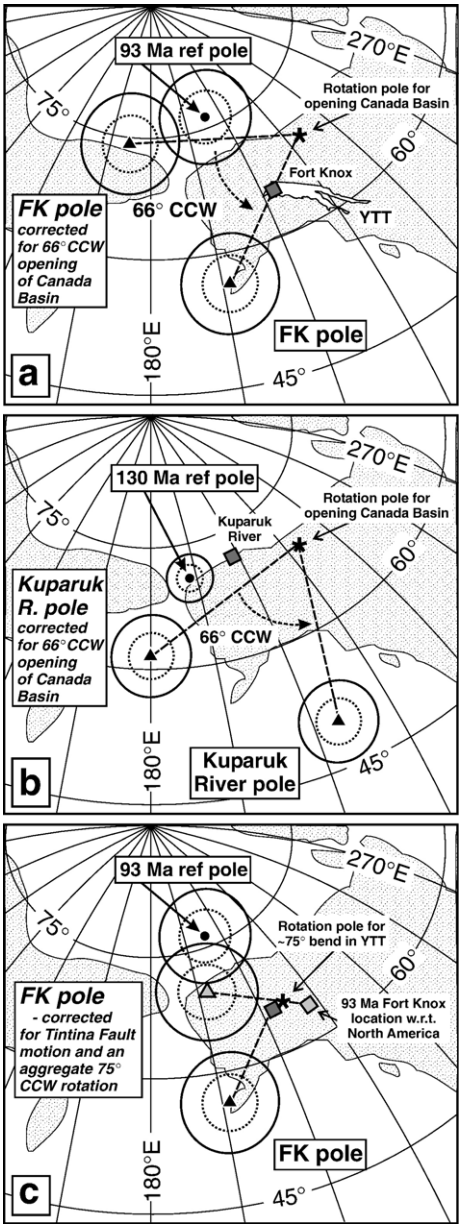


Fig. 5. (a) Location of the Fort Knox stock's paleopole position at present (FK pole) and after correction for the 66° CCW rotation of Arctic Alaska to open the Canada Basin (*FK* pole) relative to the 92.5 Ma reference pole for North America (Besse and Courtillot, 2002). The poles are surrounded by their 2σ (95%) and 1σ cones of confidence. (b) Location of the Kuparuk River Formation's paleopole at present (Kuparuk River pole) and after correction for the opening of the Canada Basin (*Kuparuk R.* pole) from Halgedahl and Jarrard (1987). (c) Location of the Fort Knox paleopole at present (FK pole), and after correction (*FK′* pole) for opening of the Canada Basin, translation on the Tintina fault, and rotation in the Alaska orocline following Plafker and Berg (1994a,b). See text for discussion.

$\gg$95% confidence that the correction on its own is clearly inadequate. There are two other aspects of this result. One is that it suggests that the majority of the rotation to open the Canada Basin occurred during the Late Cretaceous, which favours the interpretation of Lawver and Baggeroer (1983) and of McWhae (1986) rather than the Early Cretaceous interpretation of Tailleur and Brosgé (1970) and Halgedahl and Jarrard (1987). The second aspect is that the interpretation of Halgedahl and Jarrard (1987) was based on paleomagnetic results from the ~135 Ma Kuparuk River Formation, which they assumed to have acquired its ChRM at 120 Ma. Since their report, the location of the cratonic reference pole has been more precisely defined by Besse and Courtillot (2002) and is quite different from the reference paleopole that Halgedahl and Jarrard (1987) used. As a consequence, the Kuparuk River paleopole no longer provides as good paleomagnetic evidence for the ~66° CCW rotation of Arctic Alaska to open the Canada Basin (Fig. 5b). Nevertheless, correction of the Kuparuk River pole does move it much closer to the 130 Ma reference pole, indicating that the rotational opening of the Canada Basin does likely contribute substantially to the discordance between the two poles. However, even after correction the 2σ cones of confidence about the Kuparuk River and reference poles do not intersect, showing with $\gg$95% confidence that some factor other than the opening of the Canada Basin needs to be taken into account. Note also, that using a 120 Ma reference pole for the ChRM acquisition by the Kuparuk River Formation would further degrade the agreement.

In the next three sections of this paper, the fit of the Fort Knox paleomagnetic result is examined relative to the recent tectonic models for the evolution of the YTT of Plafker and Berg (1994b), Johnston (2001) and Symons et al. (2005).

5.5. *Plafker and Berg (1994b)* model

In their tectonic overview and reconstruction of Alaska, Plafker and Berg (1994b) show the Canada Basin to have opened by 100 Ma. They depict a total rotation of ~75° CCW for the Fort Knox region of the YTT between 120 and 50 Ma in response to opening of the Canada Basin, to ~450 km of dextral slip of the Tintina fault system without a significant change in paleolatitude, and to formation of the western Alaskan orocline. In their reconstruction the YTT is a distinct part of their Yukon Composite Terrane with Fairbanks as a convenient marker location. Their reconstruction considers fault displacement estimates, magmatic belt trends, structural trends, and available paleomagnetic

data. They note an aggregate rotation of ~75° CCW between the YTT in Yukon where it trends ~N50°W and in Alaska near Fairbanks where it trends ~S55°W. For the western end of the YTT in Alaska, their reconstruction contains the following elements: (1) the YTT is a linear belt at 120 Ma; (2) between 120 and 84 Ma, the Canada Basin opened as Arctic Alaska rotated ~66° CCW about an Euler pole in the Mackenzie delta, causing the Fairbanks Belt to rotate ~25° CCW about a vertical axis ~100 km east of Fairbanks; (3) between 84 and 66 Ma, ~450 km of dextral displacement occurred across the Tintina Fault zone with the Fairbanks Belt rotated a further ~10° CCW about an Euler pole ~300 km east of Fairbanks, close to the Tintina Fault; (4) between 66 and 50 Ma, compression between Eurasia and North America caused western and central Alaska to undergo a further 40±10° CCW rotation about an Euler pole ~200 km east of Fairbanks, forming the western Alaskan orocline, and causing the Fairbanks Belt to rotate a further 40±10° CCW and Baja Alaska to move northwestward by ~350 km on the dextral Denali Fault zone in the process; and, (5) between ~50 Ma and the present there were no further significant tectonic changes for the Fairbanks Belt.

In the above scenario, the Fort Knox pluton was emplaced at ~222°E, 63°N in present-day North American coordinates, and it was transported to its present location at 212.6°E, 65°N by both the ~450 km of dextral displacement on the Tintina fault and the ~75° of aggregate CCW rotation of the YTT. Note that the initial ~25° of the ~75° of the CCW rotation takes place between intrusion of the pluton at 92.5 Ma and ~84 Ma, for which two explanations appear possible. If the ~66° CCW rotation of Arctic Alaska was completed to create the Canada Basin by 92.5 Ma as some authors suggest (e.g. Tailleur and Brosgé, 1970; Halgedahl and Jarrard, 1987; Lawver et al., 2002), then active Late Cretaceous–Paleogene low-angle thrust faulting in the Brooks Range orogen (Moore et al., 1994) could have generated the ~25° of CCW rotation in the YTT. Alternatively, if the opening of the Canada Basin was not completed until after 84 Ma (e.g. Lawver and Baggeroer, 1983; Grantz et al., 1998), then either the docking and initial accretion of Arctic Alaska against the YTT or/and the subsequent Late Cretaceous–Paleocene thrust faulting in the orogen could have caused the initial ~25° of CCW rotation in the YTT.

Correcting then for the entire 75° CCW rotation of the YTT indicates that the original ChRM for the Fort Knox pluton was directed at $D=304°$, $I=84.3°$ ($\alpha_{95}=3.0°$), yielding a pole position on emplacement at 197°E, 67.4°N ($\alpha_{95}=5.9°$) (Fig. 5c). The corrections provided by applying the Plafker and Berg (1994b) tectonic model produces an excellent, but not perfect, explanation for the observed paleomagnetic discordance between the Fort Knox stock and North America. This is shown in Fig. 5c by the fact that the 1σ cone of confidence about the stock's pole is within 1° of intersecting the 1σ cone of the reference pole, showing that the two populations of directions can be drawn from one coeval population with nearly 95% confidence. Thus we conclude that this model provides an excellent explanation for the tectonic evolution of the Fairbanks Belt of the YTT after 92.5 Ma.

5.6. Ribbon continent model

The SAYBIA (Siberia–Alaska–Yukon–British ColumbIA) model suggests that the Arctic Alaska, Ruby, Dillinger and Yukon Tanana terranes, with the latter terrane extending southward through British Columbia, originally formed a linear ribbon continent that was ~8000 km long and ~500 km wide (Johnston, 2001). At about 85 Ma, SAYBIA began moving rapidly northward at ~10 cm/yr on the Kula plate. However, SAYBIA's northern end was pinned against the Chukotka Peninsula so that the ribbon continent underwent sequential oroclinal folding southward, and collapsed into its present individual terranes. Thus the southernmost YTT moved 3000±500 km northward and docked against North America by ~45 Ma. At present, the SAYBIA model leaves major questions unanswered. For example, how did the ~8000 km long ribbon continent survive as a independent plate through much of the Paleozoic and Mesozoic without being disarticulated by continental collision or rifting or transcurrent faulting? Why does the proposed collisional boundary between SAYBIA and North America record deformation events from Late Jurassic to Paleocene when the two continents did not collide until Paleocene–Eocene time? Similarly, what was the source that shed eastward-thinning Jurassic and Cretaceous sediments in the Western Canada Sedimentary Basin on the autochthonous North American margin prior to its collision with SAYBIA? Why, also, is no relict subduction zone observed in the seismic profiles? The 3000±500 km of northward displacement for the YTT also raises major questions. The estimate is obtained from the length of the unbuckled alignment of SAYBIA's terranes and the use of only three of 16 Late Cretaceous–Paleocene paleopoles now available for the YTT and IMB terranes, namely the Carmacks volcanics' (Johnston et al., 1996; Wynne et al., 1998), Churn Creek conglomerates' (Enkin et al., 2003), and Silverquick conglomerates' (Wynne et al., 1995) paleopoles. The 70 Ma Carmacks volcanics rest in scattered localities on strata of both the YTT and the IMB terranes in Yukon, and the

~89 Ma Churn Creek and ~97 Ma Silverquick conglomerates outcrop on the western margin of the IMB terranes in southern British Columbia. The salient point is that the Carmacks paleopole is discordant from a cluster of five other Late Cretaceous–Paleocene paleopoles from the YTT and from a cluster of eight other Late Cretaceous–Paleocene paleopoles from the IMB terranes (Symons et al., 2005). Similarly, the Churn Creek and Silverquick conglomerates' paleopoles are discordant from the Late Cretaceous–Paleocene cluster of eight paleopoles for the IMB that includes paleopoles from the Churn Creek volcanics and Spius volcanics in the same general region of southern British Columbia. Further, the YTT and IMB clusters of paleopoles indicate negligible and only ~1000 km of northward translation, respectively, relative to North America rather than the ~3000 km predicted by the SAYBIA model. Finally, this paleomagnetic study of the Fort Knox pluton does provide support for the SAYBIA hypothesis by recording a realistic estimate for counterclockwise rotation of the Fairbanks belt of the YTT, but it conflicts with the hypothesis by indicating negligible northward translation of the YTT after 92.5 Ma.

5.7. Baja BC "thin-skin" model

The Baja BC "thin-skin" model for the evolution of the accreted northern Canadian Cordillera (Symons et al., 2005) is consistent with earlier versions of the Baja BC model (Irving, 1985; Irving et al., 1995) in two fundamental ways. First, both models accept that the YTT has been pericratonic with respect to North America since the Early to Middle Jurassic and, second, both models agree that the IMB terranes were translated northward with respect to North America from mid-Cretaceous to mid-Eocene time. There are also three major differences. First, previous Baja BC models were based primarily on paleomagnetic paleolatitudes and thus posit northward translation distances of ~1500 to ~3500 km for the IMB terranes (e.g. Wynne et al., 1995; Irving et al., 1996). Second, these previous models have the IMB terranes docking as new crust against North America on dextral transpressional structures during the early Tertiary. Third, the variable but generally clockwise rotations recorded by the paleomagnetic data are ignored, attributed to remagnetization, or attributed to undiscussed tectonics. In contrast, the Baja BC "thin-skin" model is based primarily on paleopoles and posits a northward translation distance of 915 ± 775 km (Symons et al., 2005). Further, this model invokes thin-skin tectonics with the IMB terranes being obducted onto the YTT and underlying westward-thinning Precambrian crustal ramp of North America in early Tertiary prior to ~54 Ma. Also, in this Baja BC "thin-skin" model, the clockwise rotations observed in most paleomagnetic ChRM vectors are attributed in part to pre-accretion rotation, but also in part to thin-skinned rotation of the IMB terranes on the YTT and on the underlying Precambrian ramp from the mid-Eocene to present as shown clearly by the paleopoles for this time period (Symons et al., 2004).

The salient point is that the Baja BC "thin-skin" model (Symons et al., 2005) provides a coherent explanation for nearly all of the ~40 reliable paleopoles from Jurassic and younger rock units that are now available for the YTT and IMB terranes, excluding only the three anomalous paleopoles utilized in most of the earlier Baja BC models (e.g. Wynne et al., 1995) and in the SAYBIA model (Johnston, 2001). Finally, the Baja BC "thin-skin" model is entirely compatible with the paleomagnetic result from the Fort Knox pluton and with the interpretation of Alaskan geology provided by Plafker and Berg (1994a,b) for the YTT after 92.5 Ma.

6. Conclusions

The 92.5 Ma Fort Knox stock retains a primary thermoremanent magnetization that resides mostly in single- to pseudosingle-domain magnetite with minor pyrrhotite. The ChRM is normal in polarity, consistent with its acquisition during the Cretaceous long normal superchron from 119 to 83 Ma (Opdyke and Channell, 1996), and is directed at $D = 228.8°$, $I = 84.3°$ ($N = 18$, $k = 130$, $\alpha_{95} = 3.0°$).

The stock's paleopole is discordant relative to the coeval paleopole for North America. Its paleoinclination yields a nonsignificant poleward (northwestward) translation of 0.2° ± 6.4° (25 ± 720 km), indicating negligible displacement relative to the craton since 92.5 Ma. When combined with the few other paleolatitude estimates for the YTT in Yukon, the data indicate a nonsignificant poleward displacement of ~250 to ~450 km, suggesting that the entire YTT has been pericratonic and parautocthonous since mid-Cretaceous.

The discordance in the Fort Knox stock's paleopole shows that the pluton has been rotated by a significant 121 ± 35° CCW relative to the craton. This result agrees closely with the tectonic model for Alaska of Plafker and Berg (1994a,b) as it relates to the western end of the Fairbanks belt of the YTT. Their model implies an aggregate of ~75° CCW with ~25° CCW rotation during the final stages of the opening of the Canada Basin from 92.5 to 84 Ma, another ~10° CCW rotation as the Tintina fault underwent ~425 km of dextral displacement from

84 to ~66 Ma, and the final ~40° CCW as the western Alaskan orocline formed from ~66 to 50 Ma. The very good fit of the Fort Knox paleopole to the Plafker and Berg (1994b) tectonic model suggests that the initial interpretations made in this study are correct that the stock has been neither remagnetized nor tilted significantly since emplacement. The Fort Knox pluton's paleomagnetic result does not support tectonic models for the northern Cordillera that advocate large-scale (>1500 km) northward translation for both the YTT and IMB terranes from mid-Cretaceous to mid-Eocene; it does support models that advocate Early to Middle Jurassic accretion of the YTT to North America and Paleocene–Eocene accretion for the IMB terranes after minor (<1500 km) northward translation.

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References

Bailey, R.C., Halls, H.C., 1984. Estimate of confidence in paleomagnetic directions derived from mixed remagnetization circle and direct observational data. J. Geophys. 54, 174–182.

Bakke, A.A., 1995. The Fort Knox 'porphyry' gold deposit-structurally controlled stockwork and shear quartz vein, sulfide-poor mineralization hosted by a Late Cretaceous pluton, east-central Alaska. In: Schroeter, T.G. (Ed.), Porphyry deposits of the North-western Cordillera of North America. Can. Inst. Min. Met. Petroleum Spec., vol. 46, pp. 795–802.

Beck Jr., M.E., 1980. Paleomagnetic record of plate-margin tectonic processes along the western edge of North America. J. Geophys. Res. 85, 7115–7131.

Besse, J., Courtillot, V., 2002. Apparent and true polar wander and the geometry of the geomagnetic field over the last 200 Myr. J. Geophys. Res. 107 (B11), 6, 1–31.

Box, S.E., 1985. Early Cretaceous orogenic belt in northwestern Alaska: Internal organization, lateral extent, and tectonic interpretation. In: Howell, D.G. (Ed.), Tectonostratigraphic Terranes of the Circum-Pacific Region. Circum-Pacific Council for Energy and Mineral Resources. Earth Sci. Ser., vol. 1, pp. 137–145. Houston, TX.

Carey, S.W., 1955. The orocline concept in geotectonics. Roy. Soc. Tasmania Proc. 89, 255–288.

Carey, S.W., 1958. The tectonic approach to continental drift. Proceedings, Continental drift, a symposium. Hobart, March, 1956. Univ. Tasmania, pp. 177–355.

Cook, F.A., Clowes, R.M., Snyder, D.B., Van der Velden, A.J., Hall, K.W., Erdmer, P., Evenchick, C.A., 2004. Precambrian crust beneath the Mesozoic northern Canadian Cordillera discovered by Lithoprobe seismic reflection profiling. Tectonics 23, TC2010. doi:10.1029/2002TC001412.

Demarest Jr., H.H., 1983. Error analysis for the determination of tectonic rotation from paleomagnetic data. J. Geophys. Res. 88, 4321–4328.

Dover, J.H., 1994. Geology of part of east-central Alaska. In: Plafker, G., Berg, H.C. (Eds.), The Geology of Alaska, G1. Geol. Soc. Amer., Boulder, CO, pp. 153–204.

Douglas, T.A., Layer, P.W., Newberry, R.J., Keskinen, M.J., 2002. Geochronologic and thermobarometric constraints on the metamorphic history of the Fairbanks mining district, western Yukon–Tanana terrane, Alaska. Can. J. Earth Sci. 39, 1107–1126.

Dunlap, W.J., 2000. Nature's diffusion experiment: the cooling-rate cooling-age correlation. Geology 28, 139–142.

Enkin, R.J., Baker, J., Mahoney, J.B., Riesterer, J., 2003. Deciphering paleomagnetic inclinations: 2. Implications from Late Cretaceous strata overlapping the Insular/Intermontane Superterrane boundary in the southern Canadian Cordillera. J. Geophys. Res. 108 (B4), 1–19.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. Lond., A 217, 295–305.

Gabrielse, H., 1985. Major dextral transcurrent displacements along the Northern Rocky Mountain Trench and related lineaments in north-central British Columbia. Geol. Soc. Am. Bull. 96, 1–14.

Gabrielse, H., Monger, J.W.H., Wheeler, J.O., Yorath, C.J., 1991. Tectonic framework, Chapter 2. In: Gabrielse, H., Yorath, C.J. (Eds.), Geology of the Cordilleran Orogen in Canada. Geol. Surv. Can. Geology of Canada, vol. 4, pp. 15–28.

Gordey, S.P., Makepeace, A.J. (compilers), 2003. Yukon digital geology. Geol. Sur. Can., Open File 1749, and Yukon Geol. Sur., Open File 2003-9(D), 2 CD-ROMs.

Grantz, A., Eittreim, S.L., Dinter, D.A., 1979. Geology and tectonic development of the continental margin north of Alaska. Tectonophysics 59, 263–291.

Grantz, A., May, S.D., Taylor, P.T., Lawver, L.A., 1990. Canada Basin. In: Grantz, A., Johnson, L., Sweeney, J.F. (Eds.), The Geology of North America, L-1, The Arctic Ocean region. Geological Society of America, pp. 379–402.

Grantz, A., Clark, D.L., Phillips, R.L., Srivastava, S.P., 1998. Phanerozoic stratigraphy of Northwind Ridge, magnetic anomalies in the Canada basin, and the geometry and timing of rifting in the Amerasia basin, Arctic Ocean. Geol. Soc. Amer. Bull. 110, 801–820.

Halgedahl, S.L., Jarrard, R.D., 1987. Paleomagnetism of the Kuparuk River Formation from oriented drill core: evidence for rotation of Arctic Alaska Plate. In: Tailleur, I., Weimer, P. (Eds.), Alaskan North Slope Geology, 2, Soc. Econ. Paleontol. Minerolog., Pacific Sec., Bakersfield, pp. 581–617.

Hammer, P.T.C., Clowes, M., 2004. Accreted terranes of northwestern British Columbia, Canada: lithospheric velocity structure and tectonics. J. Geophys. Res. 109, B06305. doi:10.1029/2003JB002749.

Hansen, V.L., Dusel-Bacon, C., 1998. Structural and kinematic evolution of the Yukon–Tanana upland tectonites, east-central Alaska: a record of late Paleozoic to Mesozoic crustal assembly. Geol. Soc. Amer. Bull. 110, 211–230.

Harbert, W., Frei, L., Jarrard, R., Halgedahl, S., Engebretson, D., 1990. Paleomagnetic and plate-tectonic constraints on the evolution of

the Alaskan-eastern Siberian Arctic. In: Grantz, A., Johnson, L., Sweeney, J.F. (Eds.), The Geology of North America, L-1, The Arctic Ocean region. Geol. Soc. Am., pp. 567–592.

Hart, C.J.R., Goldfarb, R.J., Lewis, L.L., Mair, J.L., 2004. The Northern Cordilleran Mid-Cretaceous plutonic province: ilmenite/magnetite-series granitoids and intrusion-related mineralisation. Resour. Geol. 54, 253–280.

Hillhouse, J.W., Coe, R.S., 1994. Paleomagnetic data from Alaska. In: Plafker, G., Berg, H.C. (Eds.), The Geology of North America, G-1, The Geology of Alaska. Geol. Soc. Am., pp. 797–812.

Irving, E., 1985. Whence British Columbia? Nature 314, 673–674.

Irving, E., Thorkelson, D.J., Wheadon, P.M., Enkin, R.J., 1995. Paleomagnetism of the Spences Bridge Group and northward displacement of the Intermontane Belt, British Columbia: a second look. J. Geophys. Res. 100, 6057–6071.

Irving, E., Wynne, P.J., Thorkelson, D.J., Dudka, S., 1996. Large (1000 to 4000 km) northward movements of tectonic domains in the northern Cordillera, 83 to 45 Ma. J. Geophys. Res. 101, 17901–17916.

Johnston, S.T., 2001. The great Alaskan terrane wreck: reconciliation of paleomagnetic and geologic data in the northern Cordillera. Earth Planet. Sci. Lett. 6018, 1–14.

Johnston, S.T., Wynne, P.J., Francis, D., Hart, C.J.R., Enkin, R.J., Engebretson, D.C., 1996. Yellowstone in Yukon: the Late Cretaceous Carmacks Group. Geology 24, 997–1000.

Kirschvink, J.L., 1980. The least squares line and plane and the analysis of paleomagnetic data. Geophys. J. R. Astron. Soc. 62, 699–718.

Lang, J.R., Baker, T., Hart, C.J.R., Mortensen, J.K., 2000. An exploration model for intrusion-related gold systems. Soc. Econ. Geol. Newsl. 40, 1–15.

Lawver, L.A., Baggeroer, A., 1983. A note on the age of the Canada Basin. J. Alsk. Geol. Soc. 2, 57–66.

Lawver, L.A., Scotese, C.R., 1990. A review of tectonic models for the evolution of the Canada Basin. In: Grantz, A., Johnson, L., Sweeney, J.F. (Eds.), The Geology of North America, L-1, The Arctic Ocean region. Geol. Soc. Am., pp. 593–618.

Lawver, L.A., Grantz, A., Gahagan, L.M., 2002. Plate kinematic evolution of the present Arctic region since the Ordovician. In: Miller, E.L., Grantz, A., Klemperer, S.L. (Eds.), Tectonic evolution of the Bering Shelf–Chukchi Sea–Arctic Margin and Adjacent Landmasses. Geol. Soc. Am. Spec. Paper, vol. 360. Boulder, Colorado, pp. 333–358.

Lewchuk, M.T., Foucher, J., Elmore, R.D., 2004a. Paleomagnetism of the Mesozoic Asik Mountain mafic complex in northern Alaska: implications for the tectonic history of the Arctic Composite Terrane. Econ. Geol. 99, 1345–1354.

Lewchuk, M.T., Leach, D.L., Kelley, K.D., Symons, D.T.A., 2004b. Paleomagnetism of the Red Dog Zn–Pb massive sulphide deposit in the northern Alaska. Econ. Geol. 99, 1555–1567.

McCausland, P.J.A., 2002. Geotectonic motions of the Yukon–Tanana terrane, northern Cordillera: paleomagnetism and geobarometry of Mesozoic and Cenozoic intrusions. Ph.D., The Univ. Western Ontario, London, ON, Canada.

McCausland, P.J.A., Symons, D.T.A., Hart, C.J.R., Blackburn, W.H., 2001. Paleomagnetic study of the Late Cretaceous Seymour Creek stock, Yukon: minimal geotectonic motion of the Yukon–Tanana Terrane. Yukon Explor. Geol. 2000, 207–216.

McCausland, P.J.A., Symons, D.T.A., Hart, C.J.R., Blackburn, W.H., in press. Assembly of the northern Cordillera: paleomagnetic evidence for coherent, moderate Jurassic to Eocene motion of the Intermontane Belt and Yukon–Tanana terranes. In: Enkin, R., Monger, J., Haggart, J. (Eds.), Paleogeography of western North America: Constraints on latitudinal displacements, Geol. Assoc. Can. Spec. Vol. xx: xxx-xxx.

McCoy, D., Newberry, R.J., Layer, P., DiMarchi, J.J., Bakke, A., Masterman, J.S., Minehane, D.L., 1997. Plutonic-related gold deposits of interior Alaska. In: Goldfarb, R.J., Miller, L.D. (Eds.), Mineral deposits of Alaska. Econ. Geol. Mon., vol. 9, pp. 191–241.

McWhae, J.R., 1986. Tectonic history of the North Pacific, Canadian Arctic, and Spitsbergan regions since Early Cretaceous time. Am. Assoc. Pet. Geol. 70, 430–450.

Mihalynuk, M.G., Smith, M.T., Hancock, K.D., Dudka, S., 1994. Regional and economic geology of the Tulsequah River and Glacier areas (104W12 and 13), Geological fieldwork 1993. BC Ministry of Ener., Mines, and Petrol. Resour., 321–342.

Moore, T.E., Wallace, W.K., Bird, K.J., Karl, S.M., Mull, C.G., Dillon, J.T., 1994. Geology of northern Alaska. In: Plafker, G., Berg, H.C. (Eds.), The geology of Alaska. Geol. Soc. Am., Geology of North America, vol. G1. Boulder, Colorado, pp. 535–554.

Mortensen, J.K., 1992. Pre-Mid-Mesozoic tectonic evolution of the Yukon–Tanana Terrane, Yukon and Alaska. Tectonics 11 (4), 836–853.

Mortensen, J.K., Jilson, G.A., 1985. Evolution of the Yukon–Tanana terrane: evidence from southeastern Yukon Territory. Geology 13, 806–810.

Mortensen, J.K., Hart, C.J.R., Murphy, D., Heffernan, S., 2000. Temporal evolution of Early and mid-Cretaceous magmatism in the Tintina gold belt. In: Smith, M., Tucker, T. (Eds.), The Tintina gold belt: concepts, exploration, and discoveries. Yukon Chamber of Mines, Spec., vol. 2, pp. 49–58.

Murphy, J.M., Bakke, A., 1993. Low temperature thermal history of the Gilmore Dome area, Fairbanks mining district, Alaska. Can. J. Earth Sci. 30, 764–768.

Newberry, R.J., Bundtzen, T.K., Clautice, K.H., Combellick, R.A., Douglas, T., Laird, G.M., Liss, S.A., Pinney, D.S., Reifenstuhl, R.R., Stolie, D.N., 1996. Preliminary geological map of the Fairbanks mining district. Alaska Div. Geol. Geophys. Surv. Public Data File 96-16, scale 1:63,360.

Opdyke, N.D., Channell, J.E.T., 1996. Magnetic stratigraphy. Academic Press, San Diego. 346 pp.

Packer, D.R., Stone, D.B., 1972. An Alaskan Jurassic paleomagnetic pole and the Alaskan orocline. Nature 237, 25–26.

Packer, D.R., Stone, D.B., 1974. Paleomagnetism of Jurassic rocks from southern Alaska, and the tectonic implications. Can. J. Earth Sci. 11, 976–997.

Pavlis, T.L., Sisson, V.B., Foster, H.L., Nokleberg, W.J., Plafker, G., 1993. Mid Cretaceous extensional tectonics of the Yukon–Tanana terrane, Trans-Alaska Crustal Transect (TACT), east-central Alaska. Tectonics 12, 103–122.

Plafker, G., Berg, H.C., 1994a. Introduction. In: Plafker, G., Berg, H.C. (Eds.), The Geology of North America, G-1, The Geology of Alaska. Geol. Soc. Am., pp. 1–16.

Plafker, G., Berg, H.C., 1994b. Overview of the geology and tectonic evolution of Alaska. In: Plafker, G., Berg, H.C. (Eds.), The Geology of North America, G-1, The Geology of Alaska. Geol. Soc. Am., pp. 989–1021.

Robinson, M.S., Smith, T.E., Metz, P.A., 1990. Bedrock geology of the Fairbanks Mining District. Alaska, Div. Geol. Geophys. Surv., Prof. Rep. 106 (2 sheets).

Roe, J.T., Stone, D.B., 1993. Paleomagnetism of the Fairbanks basalts, interior Alaska, short notes on Alaskan geology. Prof. Rep. 113, 61–71.

Selby, D., Creaser, R.A., Hart, C.J.R, Rombach, C.S., Thompson, J.F.H., Smith, M.T., Bakke, A.A., Goldfarb, R.J., 2002. Absolute timing of sulfide and gold mineralization: a comparison of Re–Os molybdenite

and Ar–Ar mica methods from the Tintina Gold Belt, Alaska. Geology 30, 791–794.

Selby, D., Creaser, R.A., Heaman, L.M., Hart, C.J.R., 2003. Re–Os and U–Pb geochronology of the Clear Creek, Dublin Gulch, and Mactung deposits, Tombstone gold belt, Yukon, Canada: absolute timing relationships between plutonism and mineralization. Can. J. Earth Sci. 40, 1839–1852.

Snyder, D.B., Clowes, R.M., Cook, F.A., Erdmer, P., Evanchick, C.A., Van der Velden, A.J., Hall, K.W., 2002. Proterozoic prism arrests suspect terranes: insights in ancient Cordilleran margin from seismic reflection data. Geol. Soc. Amer. Today 12, 4–10.

Stone, D.B., 1989. Paleogeography and rotations of Arctic Alaska: an unresolved problem. In: Kissel, C., Laj, C. (Eds.), Paleomagnetic rotations and continental deformation. Kluwer Academic Publishers, pp. 343–364.

Symons, D.T.A., Cioppa, M.T., 2000. Crossover Plots: a useful method for plotting SIRM data in paleomagnetism. Geophys. Res. Lett. 27 (12), 1779–1782.

Symons, D.T.A., Williams, P.R., McCausland, P.J.A., Harris, M.J., Hart, C.J.R., Blackburn, W.H., 2000. Paleomagnetism and geobarometry of the Big Creek Batholith suggests that Yukon–Tanana Terrane has been a parautochthon since Early Jurassic. Tectonophysics 326, 57–72.

Symons, D.T.A., Hart, C.J.R., Harris, M.J., McCausland, P.J.A., 2004. Paleomagnetism of the Flat Creek pluton, Yukon: tectonics of the Intermontane terranes and Mackenzie Mountains. Geol. Soc. Am., Spec. Pap. 383, 187–204.

Symons, D.T.A., Harris, M.J., McCausland, P.J.A., Blackburn, W.H., Hart, C.J.R., 2005. Mesozoic–Cenozoic paleomagnetism of the Intermontane and Yukon–Tanana Terranes, Canadian Cordillera. Can. J. Earth Sci. (June Issue).

Tailleur, I.L., 1969. Rifting speculation on the geology of Alaska's North Slope. Oil Gas J. 67, 128–130.

Tailleur, I.L., Brosgé, W.P., 1970. Tectonic history of northern Alaska. In: Adkison, W.L., Brosge, W.P. (Eds.), Proceedings of the geological seminar on the North Slope of Alaska. Am. Assoc. Petroleum Geol., Pacific Sec., pp. E1–E19.

Tempelman-Kluit, D.J., 1976. The Yukon crystalline terrane: enigma in the Canadian Cordillera. Geol. Soc. Amer. Bull. 87, 1343–1357.

Thompson, J.F.H., Sillitoe, R.H., Baker, T., Lang, J.R., Mortensen, J.K., 1999. Intrusion-related gold deposits associated with tungsten–tin provinces. Miner. Depos. 34, 323–334.

Witte, W.K., 1982. Paleomagnetism and paleogeography of Cretaceous northern Alaska. MS Thesis, Univ. Alaska, Fairbanks, 146 p.

Witte, W.K., Stone, D.B., Mull, C.G., 1987. Paleomagnetism, paleobotany, and paleogeography of the Arctic Slope, Alaska. In: Tailleur, I., Weimer, P. (Eds.), Alaskan North Slope Geology, 2, Soc. Econ. Paleontol. Minerol., Pacific Sec., Bakersfield, pp. 571–579.

Wynne, P.J., Irving, E., Maxson, K.L., Kleinspehn, 1995. Paleomagnetism of the Upper Cretaceous strata of Mount Tatlow: evidence for 3000 km of northward displacement of the eastern Coast Belt, British Columbia. J. Geophys. Res. 100, 6073–6091.

Wynne, P.J., Enkin, R.J., Baker, J., Johnston, S.T., Hart, C.J.R., 1998. The big flush: paleomagnetic signature of a 70 Ma regional hydrothermal event in displaced rocks of the northern Canadian Cordillera. Can. J. Earth Sci. 35, 657–671.